

[New issue](#)

Tips for Linux users #64

[Open](#)

doegox opened on Oct 27, 2016

Last edited by doegox ▾

Contributor



As there is not really other place of discussion and my notes are probably not mature enough for inclusion in the official docs, I'm sharing a few tips here and hoping to gather yours as well!

I'm using a Debian, details may vary with other distros.

Setup

First my `/etc/udev/rules.d/98-ChameleonMini.rules` to operate *and* flash the chameleon as regular user and stop being annoyed by pesky ModemManager:

```
# Rule for ChameleonMini RFID Research tool
ATTRS{product}=="Chameleon-Mini", SUBSYSTEMS=="usb", ATTRS{idVendor}=="16d0",
ATTRS{idProduct}=="04b2", GROUP="users", MODE="0666", SYMLINK+="chameleon",
ENV{ID_MM_DEVICE_IGNORE}="1"
# DFU mode
SUBSYSTEMS=="usb", ATTRS{idVendor}=="03eb", ATTRS{idProduct}=="2fde", GROUP="users", MODE="0666"
```



Minicom

As terminal, minicom can do the job, but a few tweaks help a bit, with this file as `/etc/minicom/minirc.chameleon` or `~/.minirc.chameleon`:

```
pu port                /dev/chameleon
pu localecho           Yes
pu linewrap            Yes
# disable other transports than xmodem
pu pname1              YUNYY
pu pname2              YUNYY
pu pname4              NDNY
```



```
pu pname5      NDNYN
pu pname7      YUYNN
pu pname8      NDYNN
pu pname9      YUNYN
```

Then invoke it simply with:

```
minicom chameleon
```



xmodem transfers are possibles.

Screen

Usually I like `screen` as simple terminal but there is no easy way to activate local echo (see [this discussion](#))

Socat

So the alternative to get a simple terminal is to use socat.

A nice add-on is to use its readline support to get history, ctrl-r etc. Unfortunately Debian and Ubuntu versions compiled it without readline support, so if you want to have it you've to recompile the package by yourself (see [notes](#)) or use the original sources.

But getting history in your chameleon terminal is so great!

cham

```
#!/bin/bash
socat READLINE,history=$HOME/.chameleon_history /dev/chameleon,crlf
```



File transfer: Download

cham-down

```
#!/bin/bash
# Usage:
# $ cham-down [filename [slot_nr]]
# default is to dump current slot to dump.bin
[ ! -z $2 ] && echo -e "SETTING=$2\r" >/dev/chameleon && read foo </dev/chameleon
socat SYSTEM:"echo \"DOWNLOAD\r\" ;read foo; rx ${1:-dump.bin}" /dev/chameleon
```



File transfer: Upload

cham-up

```
#!/bin/bash
# Usage:
# cham-up [filename [slot_nr]]
# default is to upload dump.bin into current slot
[ -f "${1:-dump.bin}" ] || exit 1
[ ! -z $2 ] && echo -e "SETTING=$2\r" >/dev/chameleon && read foo </dev/chameleon
echo -e "UPLOAD\r" > /dev/chameleon
socat EXEC:"sx ${1:-dump.bin}" /dev/chameleon
```



Flashing

Here is my little script to compile, jump to bootloader, flash and jump back to normal mode in one go without having to unplug/plug the chameleon. This requires to install also the package `dfu-programmer`.

cham-flash

```
#!/bin/bash
cd /path/to/ChameleonMini/Firmware/Chameleon-Mini
make || exit 1
# Jump to bootloader mode
echo -e "\rUPGRADE\r" > /dev/chameleon && sleep 1
make program
# Jump to normal mode
dfu-programmer atxmega128a4u start
```



Doc

I also created a convenient

cham-man

```
x-www-browser /path/to/ChameleonMini/Doc/Doxygen/html/index.html &
```



Your turn

Any other tips to share on your side?



11



2



4



zenroth1752 on Oct 27, 2016



You are star - thank you



zenroth1752 on Feb 6, 2017



This tool allows the MFD dump to be in human readable format - <https://github.com/zhovner/mfdread>



Peterthegreat mentioned this [on Jul 30, 2017](#)

[setting & Dump Mifare Classic 4K #128](#)

fptrs mentioned this [on Mar 4, 2020](#)

[Chameleon working in windows but not linux #258](#)

Add a comment

Write

Preview



Use Markdown to format your comment

Paste, drop, or click to add files

Close issue



Comment

Metadata

Assignees

No one assigned

Labels

No labels

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

[Customize](#)

 [Unsubscribe](#)

You're receiving notifications because you're subscribed to this thread.

Participants



 [Open in GitHub Copilot app](#)

 [Give feedback](#)